

Latency-Bounded Reliability Oriented Rateless Codes for Multi-Connectivity Networks Toward URLLC: A Knowledge-Guided RL Framework and Algorithm Design

Zhaoyu Wang, Yusun Fu, *Member, IEEE*, and Weiwu Yan

Abstract—Toward ultra-reliable low-latency communications (URLLC) in Industrial Internet of Things (IIoT) scenarios, improving the latency-bounded reliability of rateless codes (RC) remains a significant challenge. RC, by exploiting multiple transmission paths, provide a promising approach to improve reliability while reducing end-to-end latency. However, in multi-connectivity (MC) networks, LT-code optimization faces intensified environment dynamics and difficult cross-path credit assignment, and existing schemes relying on static degree distributions are ill-suited for short code lengths and heterogeneous link conditions. To address these challenges, we propose MCRC, a knowledge-guided reinforcement learning (RL) framework for degree distribution optimization of multi-connectivity RC. To the best of our knowledge, we establish the first latency-aware Markov decision process (MDP) formulation for multi-path parallel rateless transmission, derive an upper bound on the multi-path joint decoding failure probability, and analytically decompose the joint return into per-path trajectory terms, providing a principled basis for cross-path credit assignment and reward design. We further develop a training strategy that combines expert imitation with curriculum-style cascaded transfer: the policy is first trained at a small code length and then warm-started stage by stage along a doubling code-length ladder, where each converged policy is analytically extended by a soliton-referenced log-degree interpolation operator and serves as the teacher of the next stage, reducing training cost by approximately $3\times$ without compromising final performance. Simulation results demonstrate that MCRC significantly outperforms classical degree distributions and single-connectivity RL-based approaches in terms of decoding success rate, end-to-end latency, and transmission overhead, providing effective support for URLLC-oriented IIoT deployments.

Index Terms—Rateless codes, Multi-connectivity, Reinforcement learning, Latency-Bounded Reliability, Degree distribution, URLLC, Imitation learning, Curriculum learning, Credit assignment.

I. INTRODUCTION

This work was supported in part by the Key Research and Development Program of Ningbo under Grant 2024Z111 and in part by the National Natural Science Foundation of China under Grant 62273233. (Corresponding author: Yusun Fu.)

Z. Wang is with the School of Automation and Intelligent Sensing, Shanghai Jiao Tong University, Shanghai 200240, China, and also with the Ningbo Artificial Intelligence Institute, Shanghai Jiao Tong University, Ningbo 315000, China (e-mail: zhihuiuzz@sjtu.edu.cn).

Y. Fu and W. Yan are with the School of Automation and Intelligent Sensing, the Key Laboratory of System Control and Information Processing, Ministry of Education of China, and the Shanghai Key Laboratory of Perception and Control in Industrial Network Systems, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: fu_yusun@sina.com; yan-wwsjtu@sjtu.edu.cn).

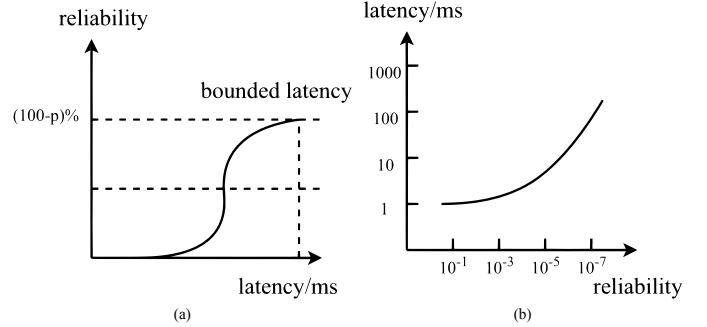


Fig. 1: The relationship between reliability and latency. (a) defines reliability under bounded latency, while (b) illustrates the trade-off between latency and reliability.

IN the Industrial Internet of Things (IIoT), ultra-reliable and low-latency communications (URLLC) have become a critical enabler for real-time industrial control, with stringent requirements on packet error rate and end-to-end latency [1]–[3]. To support these requirements, improvements at the coding layer are essential, yet the reliability-latency trade-off poses a fundamental challenge: increasing coding redundancy improves reliability but incurs additional delay, while reducing latency by limiting redundancy may compromise decoding success, as illustrated in Fig. 1 [4], [5].

Multi-connectivity (MC), which allows a terminal to simultaneously transmit data over multiple independent paths, has emerged as a promising approach to address this challenge by exploiting path diversity [6]–[8]. However, in practical IIoT deployments, heterogeneous links exhibit highly diverse latency, jitter, and channel quality [9]. For instance, a terminal may simultaneously utilize a low-latency but unstable 5G link and a more stable yet significantly slower Wi-Fi link. Under strict latency constraints, the system cannot arbitrarily wait for delayed packets from slower paths, and the resulting head-of-line blocking or forced early termination of decoding can offset the expected reliability gains [8], [10]. Simply increasing the number of parallel links does not guarantee URLLC performance without intelligent coordination mechanisms.

Rateless codes offer a natural solution to this coordination problem. Since each path independently accumulates encoded symbols and attempts belief propagation (BP) decoding without requiring cross-path coordination [11], [12], rateless codes enable multiple independent decoding attempts in parallel across paths, improving overall reliability without

complex packet reordering or link-layer retransmission [13], [14]. However, the performance of rateless codes, particularly Luby transform (LT) codes [15], heavily depends on the degree distribution, which simultaneously governs encoding overhead, decoding complexity, and success probability [16]. Classical degree distributions such as the Robust Soliton Distribution (RSD) are designed for asymptotic regimes and often perform poorly under the short code lengths and latency-bounded requirements typical of URLLC [17]–[19]. Moreover, static degree distributions cannot adapt to the heterogeneous channel conditions across multiple paths.

Existing optimization approaches, including empirical tuning [20], analytical methods [19], [21], [22], and machine learning-based techniques [23]–[27], have demonstrated improvements in degree distribution design. However, these studies predominantly focus on single-path systems [23]–[27] or non-MC latency analysis [28], without addressing the latency-reliability trade-off inherent in MC-URLLC scenarios.

More fundamentally, extending degree distribution optimization from single-path to MC scenarios is not a straightforward generalization, and LT-code optimization in MC settings faces qualitatively new challenges. First, the environment dynamics are substantially intensified: heterogeneous erasure processes evolve independently on each path, and the transmission outcome is determined by the coupled evolution of M parallel belief-propagation decoders, which enlarges the stochasticity that any optimizer must cope with. Second, credit assignment across paths becomes difficult: successful recovery is a single joint event over all paths, and it is unclear how this joint outcome should be attributed to the encoding decisions carried by individual paths, which obscures the learning signal. To date, RL-based optimization of rateless codes has not been systematically investigated in MC scenarios.

A. Contributions

To address the limitations of existing degree distribution designs under latency-bounded MC scenarios, this work proposes a reinforcement learning (RL)-based optimization framework for rateless codes. The main contributions are summarized as follows:

- **Latency-Aware MDP Modeling and Theoretical Analysis of Multi-Path Parallel Rateless Transmission:** To the best of our knowledge, this is the first work that establishes a latency-aware Markov decision process (MDP) formulation for degree distribution optimization in multi-path parallel transmission scenarios, with a reward function that jointly captures decoding success rate and end-to-end latency in heterogeneous MC systems. On top of this model, the ripple evolution analysis is extended to independent multi-path reception, and an upper bound on the multi-path joint decoding failure probability is derived (Proposition 1), which explicitly couples the latency budget, the per-path erasure probabilities, and the degree distribution, establishing that path diversity multiplicatively reduces decoding failure under a latency-constrained symbol budget. Furthermore, the joint return is analytically decomposed into per-path trajectory terms

(Proposition 2), providing a principled basis for cross-path credit assignment and for the sparse joint reward design.

- **Algorithm Design of Latency-Bounded Reliability Oriented Degree Distribution for Multi-Connectivity Rateless Codes:** In MC scenarios, LT-code optimization faces intensified environment dynamics and difficult cross-path credit assignment, and existing studies have not systematically explored RL-based optimization in such settings. Guided by the theoretical analysis, an RL-based degree distribution optimization algorithm, termed MCRC, is designed with a state space consisting of the source block length k , and learns degree distributions that maintain ripple stability across heterogeneous paths within the latency budget. This enables path-diversity-exploiting adaptive encoding that static distributions cannot achieve, maximizing latency-bounded reliability across varying code lengths and channel conditions.
- **Expert Imitation Plus Curriculum-Style Cascaded Transfer Training:** A knowledge-guided training strategy is designed that combines expert imitation with cascaded transfer across code lengths. The policy is first trained at the smallest code length ($k = 16$), where the ideal soliton distribution (ISD), a theoretically grounded classical degree distribution, serves as the expert for imitation learning (IL); each converged policy is then analytically extended to twice the code length by a soliton-referenced log-degree interpolation operator and serves as the *teacher* of the next stage ($16 \rightarrow 32 \rightarrow 64 \rightarrow \dots$) through KL-based distillation and weight inheritance. This curriculum-style cascade lets every stage start from a near-optimal initialization instead of random exploration, reducing the overall training cost by approximately $3\times$ without compromising final performance.

II. RELATED WORK

Table I summarizes the key characteristics of the reviewed literature and highlights the novelty of this work. Below we discuss the two main streams of related research.

A. Degree Distribution Optimization for LT Codes

The performance of LT codes largely depends on the degree distribution, which governs encoding overhead, decoding complexity, and success probability [35]. Classical distributions such as the RSD [15] are designed for asymptotic code lengths and often perform poorly in practical short-code scenarios [17]–[19]. Research on degree distribution optimization can be broadly classified into three categories: (i) empirical parameter tuning, such as the Poisson Robust Soliton Distribution (PRS) [20]; (ii) analytical approaches based on ripple evolution or decoding set analysis [16], [18], [19], [21]; and (iii) machine learning-based methods that dynamically search for numerically optimal distributions [23]–[27]. While these approaches have demonstrated improvements, they predominantly target single-path systems and optimize overhead or decoding performance without considering latency-bounded constraints or heterogeneous channel conditions. In particular,

TABLE I: Summary of the Characteristics of the Reviewed Articles in Related Work and the Novelty and Contributions of This Article

Paper(s)	Field of Study	Main Contributions	Our Novelty
[15]	Rateless codes schemes	Introduced LT codes as the first practical realization of fountain codes, enabling efficient rateless transmission over erasure channels.	• Focus on improving the latency-bounded reliability of rateless codes. • Provide an optimization framework that can be extended to various rateless codes schemes.
[12]		Proposed Raptor codes, which combine a pre-code with LT coding to significantly reduce encoding and decoding complexity.	
[11]		Analyzed fountain codes and their information-theoretic properties for rateless transmission over erasure channels.	
[23]	Degree distribution optimization for LT codes	Applied deep reinforcement learning to optimize LT code degree distributions for improved decoding performance.	• Optimize degree distribution under multi-path channel conditions for latency-bounded reliability, going beyond single-path optimization. • Enable adaptive coding behavior that dynamically responds to heterogeneous link conditions in MC systems.
[21]		Designed and analyzed LT codes with decreasing ripple size to improve decoding reliability.	
[20]		Proposed Poisson Robust Soliton Distribution to improve LT code performance through heuristic parameter design.	
[18], [19]	Analytical degree distribution design for rateless codes	Improved finite-blocklength rateless-code design through lower-complexity degree distribution optimization and ripple-oriented backward design.	• Directly address practical engineering requirements by optimizing latency-bounded reliability. • Focus on maximizing latency-bounded decoding reliability.
[28]		Analyzed age-of-information and reliability performance of rateless codes with different block lengths over erasure channels.	
[25]	Applications of rateless codes	Investigated intelligent degree distribution of rateless codes for reliable communication in industrial automation scenarios.	• Focus on URLLC scenarios and analyze the latency-bounded reliability of rateless codes.
[29]		Designed fountain codes for reliable and deterministic packet transmission in industrial cloud control systems.	
[30]	URLLC and reliability analysis	Investigated ultra-reliable and low-latency wireless communication from the perspective of tail risk and scalability.	• Integrate rateless codes with MC communication systems. • Propose a practical optimization method for improving reliability under multi-link transmission.
[2]		Proposed radio resource management strategies for ultra-reliable and low-latency communications.	
[3]		Surveyed ultra-low latency network standards including IEEE TSN and IETF DetNet for 5G systems.	
[31]	RL-based optimization methods for communication systems	Surveyed applications of deep reinforcement learning in communications and networking.	• Introduce RL to adaptively optimize transmission strategies in MC systems. • Enable dynamic and data-driven optimization under varying channel conditions.
[32]		Provided a comprehensive tutorial on single- and multi-agent DRL for AI-enabled wireless network optimization.	
[24]		Proposed an RL-based framework for optimizing degree distribution and feedback for short LT codes.	
[33], [34]		Applied DRL for packet duplication and power allocation in MC URLLC networks.	

RL-based degree distribution optimization has so far been confined to single-connectivity settings; a systematic treatment in MC systems—where the environment dynamics are intensified and the learning signal must be attributed across parallel paths—is still missing.

B. Rateless Codes in Industrial and Multi-Path Scenarios

Several studies have applied rateless codes to industrial communication scenarios. Reference [29] designs fountain codes for reliable and deterministic packet transmission in industrial cloud control, and [28] analyzes the latency-reliability trade-off of rateless codes over erasure channels in the context of age of information [36]. Reference [25] investigates intelligent degree distribution for industrial automation, while [37] applies Raptor codes to M2M streaming to exploit channel

diversity. Recent work on MC has explored packet duplication [8], [33] and cooperative resource allocation [34] for URLLC. However, these approaches generally rely on static scheduling or single-path optimization and lack a systematic methodology for optimizing coding parameters for latency-bounded reliability in MC environments. As highlighted in Table I, none of the existing works jointly address latency-bounded reliability optimization of rateless codes in MC systems, which is the central contribution of this paper.

III. BACKGROUND

A. LT Codes

LT codes are the first practical realization of rateless codes, designed for reliable transmission over erasure channels without requiring feedback [14], [15]. A source block of k

symbols, $\mathbf{S} = \{s_1, s_2, \dots, s_k\}$, is encoded into an essentially unlimited stream of coded symbols. Each coded symbol is generated by randomly selecting a degree d according to a predefined degree distribution $\Omega(d)$, and XOR-ing d randomly chosen source symbols. The decoding complexity scales with the average degree $\bar{d} = \sum_{d=1}^{D_{\max}} d \cdot \Omega(d)$, making the choice of $\Omega(d)$ critical to both overhead and computational cost.

A foundational degree distribution is the Ideal Soliton Distribution (ISD), defined as

$$\rho(d) = \begin{cases} \frac{1}{k}, & d = 1, \\ \frac{1}{d(d-1)}, & d = 2, 3, \dots, k, \end{cases} \quad (1)$$

which is designed so that on average exactly one coded symbol is reduced to degree-1 at each decoding step. In practice, ISD is often augmented with additional robustness parameters to form the RSD [15].

The decoding process employs belief propagation (BP) [35], [38], which iteratively recovers source symbols. At each BP iteration, a degree-1 coded symbol (or a coded symbol that has been reduced to degree-1) reveals a source symbol, which is then used to reduce the degree of other coded symbols. The set of coded symbols with degree exactly one at any point during decoding is called the *ripple*. The ripple size r_t at decoding step t evolves as

$$r_{t+1} = r_t - 1 + Q_t, \quad (2)$$

where Q_t denotes the number of new coded symbols that are reduced to degree-1 when the source symbol decoded at step t is removed from the encoding graph. BP decoding succeeds if and only if the ripple never reaches zero before all k source symbols are recovered, i.e., $r_t > 0$ for all $t \in \{1, \dots, k\}$. Consequently, maintaining a healthy ripple throughout the decoding process is the key objective of degree distribution design [21].

Since the decoding success probability $P_{\text{success}}(n, k, \Omega)$ depends on Ω through the stochastic ripple evolution process, it does not admit a tractable closed-form expression and is typically evaluated via simulation [11], [12]. This analytical intractability motivates the use of data-driven optimization approaches.

B. Reinforcement Learning

RL learns decision-making policies through interaction with the environment [39]. An RL problem is typically modeled as a Markov Decision Process (MDP), defined by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $\mathcal{P}$ the state transition probability, R the reward function, and $\gamma \in [0, 1]$ the discount factor. The agent observes a state s_t , selects an action a_t , transitions to s_{t+1} according to $\mathcal{P}$, and receives a reward R_t . Through repeated interaction, it learns a policy $\pi(a|s)$ that maximizes the expected cumulative reward using algorithms such as policy gradients [40], [41] or actor-critic methods [42], [43].

C. Multi-Connectivity

MC is a communication mechanism that enables a user device to simultaneously utilize multiple paths or different

radio access technologies (RATs) for data transmission [6]–[8]. According to 3GPP TR 23.793 [44], five traffic steering modes are defined for MC, as illustrated in Fig. 2:

- **Active-Standby:** All traffic is sent over a single designated *active* path, while the remaining path serves as a *standby*. Traffic is seamlessly switched from the active to the standby path when the active path becomes unavailable, and switched back once connectivity is restored, providing enhanced session continuity with minimal signaling overhead.
- **Priority-based:** Each path is assigned a priority level. All traffic is directed to the high-priority path under normal conditions. When congestion occurs on the high-priority path, overflow traffic is offloaded to the low-priority path; if the high-priority path becomes entirely unavailable, all traffic is switched to the alternative path, enabling both continuity and increased bandwidth utilization.
- **Best-Access:** Similar to the priority-based mode, but the preferred path is not statically predefined. Instead, the system estimates path performance (e.g., round-trip time) in real time and dynamically steers traffic to the path offering the best current quality, achieving improved performance at the cost of more complex path monitoring.
- **Redundant:** Identical data is transmitted simultaneously over all available paths to maximize reliability. The receiver detects and discards duplicate packets, accepting whichever copy arrives first. This mode achieves very low packet error rates and strong session continuity, at the expense of increased bandwidth consumption.
- **Load-Balance:** Traffic is split across paths according to configurable weight factors (e.g., 50%/50% or 80%/20%). Each path carries a proportional share of the total traffic, providing bandwidth aggregation without requiring real-time performance monitoring of individual paths.

Among these modes, the *redundant* mode is particularly relevant to URLLC, where reliability is the primary objective. In this work, we focus on the redundant steering mode: encoded symbols generated by rateless codes are transmitted over M parallel paths simultaneously, and each path independently delivers symbols to the receiver for decoding. Since each path independently performs BP decoding on its own received symbol set, this architecture exploits path diversity through parallel independent decoding attempts, maximizing the probability that at least one path succeeds within the latency-bounded constraints.

D. Ripple Evolution Under Multi-Path Reception

We now analyze multi-path decoding failure from the ripple-evolution perspective. By extending the single-path ripple analysis to independent multi-path reception, we show that path diversity multiplicatively reduces BP decoding failure probability, establishing the ripple-level superiority of multi-path reception over single-path regardless of degree distribution.

In the single-path case, Sørensen *et al.* [21] and Huang *et al.* [19] derived the release probability $q(d, L, R)$, i.e., the probability that a coded symbol of original degree d is

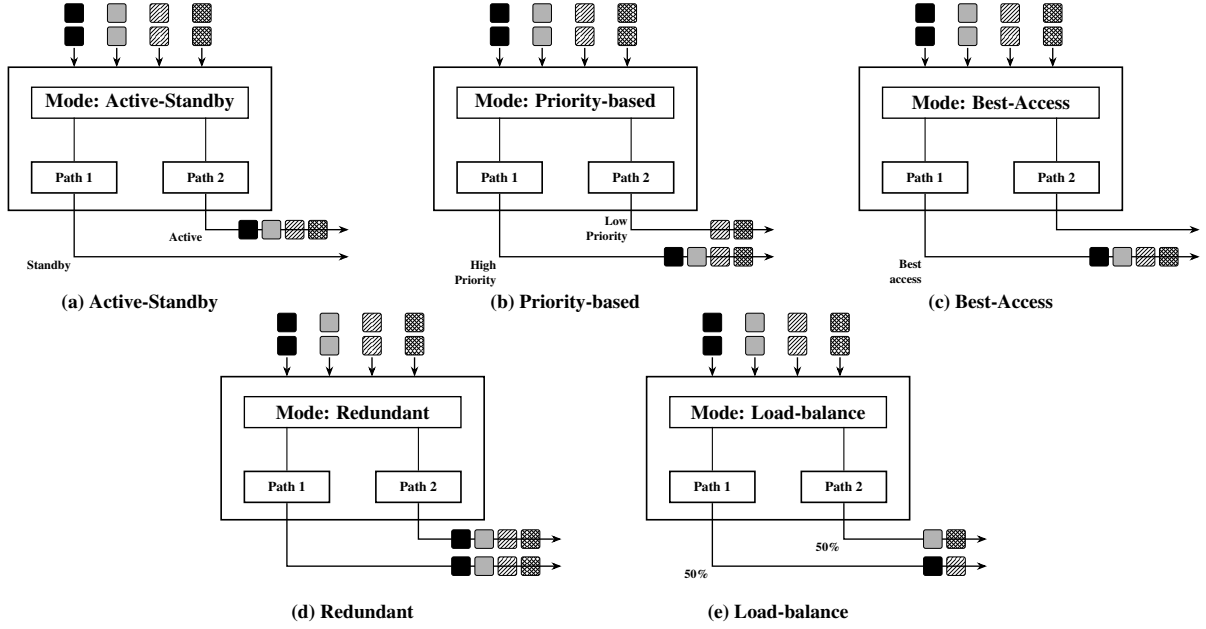


Fig. 2: Five traffic steering modes for multi-connectivity defined in 3GPP TR 23.793 [44].

released and added to the ripple when L source symbols remain unprocessed and the current ripple size is R :

$$q(d, L, R) = \frac{d(d-1)(L-R+1) \prod_{i=0}^{d-3} (k-(L+1)-i)}{\prod_{i=0}^{d-1} (k-i)}, \quad (3)$$

for $d = 2, \dots, k$, $R = 1, \dots, k-d+1$, $L = R, \dots, k-d+1$, with $q(1, k, 0) = 1$ and $q(d, L, R) = 0$ otherwise. The expected number of symbols added to the ripple at decoding step t (when $L = k-t$ symbols remain) from n received coded symbols is

$$\mathbb{E}[A(L)] = \sum_{d=2}^k n \cdot \Omega(d) \cdot q(d, L, R), \quad (4)$$

and the expected ripple evolution follows

$$\mathbb{E}[R(L-1)] = \mathbb{E}[R(L)] - 1 + \mathbb{E}[A(L)]. \quad (5)$$

In our MC system with M independent paths operating in redundant mode, each path independently generates its own stream of coded symbols by sampling from the shared degree distribution Ω , transmitting n coded symbols in total. Path j independently receives each of its n transmitted symbols with probability $1 - \varepsilon_j$, yielding an expected $n_j^{\text{recv}} = n(1 - \varepsilon_j)$ received symbols. Since each path maintains its own BP decoder, the ripple evolution on path j follows the single-path model with n replaced by $n(1 - \varepsilon_j)$:

$$\mathbb{E}[A^{(j)}(L)] = \sum_{d=2}^k n(1 - \varepsilon_j) \cdot \Omega(d) \cdot q(d, L, R^{(j)}), \quad (6)$$

$$\mathbb{E}[R^{(j)}(L-1)] = \mathbb{E}[R^{(j)}(L)] - 1 + \mathbb{E}[A^{(j)}(L)], \quad (7)$$

where $R^{(j)}$ denotes the ripple size on path j . BP decoding on path j succeeds if and only if $R^{(j)}(L) > 0$ for all $L \in \{1, \dots, k-1\}$. Since the M paths decode independently, the overall decoding failure probability is

$$P_{\text{fail}}^{\text{MP}} = \prod_{j=1}^M P_{\text{fail}}^{(j)}, \quad (8)$$

where $P_{\text{fail}}^{(j)} = \Pr[\exists L : R^{(j)}(L) = 0]$ is the decoding failure probability on path j .

The multiplicative structure in (8) can be sharpened into an explicit upper bound that couples the latency budget, the per-path channel qualities, and the degree distribution. Let $P_{\text{BP}}(m, k, \Omega)$ denote the probability that BP decoding fails to recover all k source symbols from m received coded symbols whose degrees are drawn i.i.d. from Ω ; since additional received symbols can only assist BP decoding, $P_{\text{BP}}(m, k, \Omega)$ is non-increasing in m , and it can be evaluated through the ripple recursion (6)–(7).

Proposition 1 (Multi-path joint decoding failure probability bound): Suppose the latency budget allows each path to transmit n coded symbols, and path j receives each symbol independently with probability $1 - \varepsilon_j$. Then, for any thresholds m_j with $k \leq m_j < n(1 - \varepsilon_j)$, $j = 1, \dots, M$,

$$P_{\text{fail}}^{\text{MP}} \leq \prod_{j=1}^M \left[e^{-\frac{2(n(1-\varepsilon_j)-m_j)^2}{n}} + P_{\text{BP}}(m_j, k, \Omega) \right]. \quad (9)$$

Proof: Let N_j denote the number of symbols received on path j , so that $N_j \sim \text{Binomial}(n, 1 - \varepsilon_j)$. Decoding failure on path j implies either a reception deficit $\{N_j < m_j\}$ or a BP failure despite sufficient reception $\{N_j \geq m_j, \text{BP fails}\}$. By the union bound and the monotonicity of P_{BP} in its first argument,

$$P_{\text{fail}}^{(j)} \leq \Pr[N_j < m_j] + P_{\text{BP}}(m_j, k, \Omega). \quad (10)$$

For $m_j < n(1 - \varepsilon_j)$, Hoeffding's inequality applied to the binomial lower tail gives

$$\Pr[N_j < m_j] \leq \exp\left(-\frac{2(n(1-\varepsilon_j)-m_j)^2}{n}\right). \quad (11)$$

Substituting (10) and (11) into (8), which holds by the independence of the M decoders, yields (9). ■

Remark 1: The bound (9) decomposes each path's failure into a *channel term*, which decays exponentially in the latency budget through the transmittable symbol count n , and a *coding term* $P_{BP}(m_j, k, \Omega)$, which is the only component that the degree distribution can influence. It also makes precise when path diversity pays off: the multiplicative gain materializes only if every bracketed factor stays below one, i.e., only if Ω keeps even the weakest path individually viable within its reduced effective symbol budget $n(1 - \varepsilon_j)$.

This analysis reveals three key insights. First, for a path with higher erasure probability ε_j , the effective number of received symbols $n(1 - \varepsilon_j)$ is smaller, leading to fewer ripple additions $\mathbb{E}[A^{(j)}(L)]$ and a higher risk of ripple depletion ($R^{(j)}(L) \rightarrow 0$). Second, multi-path diversity provides a multiplicative reduction in failure probability through (8) and (9): even if individual paths have moderate failure rates, the overall system achieves high reliability, provided that no path is left effectively unusable. Third, under the latency bound $\tau_{\max}$, the number of transmittable symbols n is limited, which inflates both terms of the bound and constrains $\mathbb{E}[A^{(j)}(L)]$ for all paths. Since the coding term is the operative lever, this motivates the use of RL to learn degree distributions $\Omega(d)$ that maximize $\mathbb{E}[A^{(j)}(L)]$ across heterogeneous paths within the latency budget.

IV. PROPOSED MCRC FRAMEWORK AND ALGORITHM DESIGN

This section presents the proposed MCRC framework and algorithm design. We first formulate the optimization problem, then describe the system model and MCRC framework, present the MCRC algorithm design and the cross-path credit assignment analysis, and finally detail the knowledge-guided training procedure with expert imitation and cascaded curriculum transfer.

A. Performance Metrics and Problem Formulation

To formally quantify the benefits of multi-path transmission, we first analyze reliability at the link level. Consider a system with M independent paths indexed by $j \in \{1, 2, \dots, M\}$. Let ε_j denote the packet erasure probability of path j . The link-level packet loss probability of path j is

$$P_{\text{link}}^{(j)} = \varepsilon_j. \quad (12)$$

When all M paths are used simultaneously in the redundant mode, the overall link-level transmission failure probability is [8]:

$$P_{\text{link}} = \prod_{j=1}^M P_{\text{link}}^{(j)} = \prod_{j=1}^M \varepsilon_j. \quad (13)$$

From the channel perspective, this multiplicative structure directly establishes the advantage of redundant multi-path

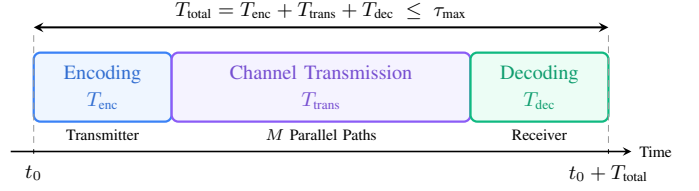


Fig. 3: Illustration of end-to-end latency components.

transmission: the joint link-level failure probability is the product of individual path erasure rates, so multi-path is inherently superior to any single-path configuration in terms of raw channel reliability, regardless of the coding strategy employed.

We define three performance metrics for evaluation:

- **End-to-end latency (T):** the time elapsed from symbol generation at the transmitter to successful decoding at the receiver.
- **Decoding success probability (P_{success}):** the probability that the source block is successfully decoded under given channel conditions within the latency bound $\tau_{\max}$.
- **Transmission overhead (O):** the number of coded symbols transmitted by the first path to successfully decode, relative to the source block size.

The end-to-end latency consists of three components:

$$T_{\text{total}} = T_{\text{enc}} + T_{\text{trans}} + T_{\text{dec}}, \quad (14)$$

where T_{enc} and T_{dec} denote the encoding and decoding computation times, respectively, and T_{trans} represents the transmission time over the wireless channel, as illustrated in Fig. 3.

The transmission overhead is defined as

$$O = \frac{n - k}{k}, \quad (15)$$

where n is the number of coded symbols transmitted on the earliest path to successfully decode and k is the number of source symbols. This per-path coding overhead directly reflects how efficiently the degree distribution enables successful decoding, and is consistent with the overhead constraint used in the reward function (20). The expected number of successfully received symbols across M paths is

$$\mathbb{E}[n_{\text{recv}}] = \sum_{j=1}^M n_j(1 - \varepsilon_j), \quad (16)$$

where n_j is the number of coded symbols transmitted over path j , ε_j is the erasure probability of path j , and M is the total number of parallel paths.

Based on these definitions, the optimization problem is formulated as

$$\begin{aligned} \max_{\Omega} \quad & P_{\text{success}}(\mathbb{E}[n_{\text{recv}}], k, \Omega) \\ \text{s.t.} \quad & T_{\text{total}} \leq \tau_{\max}, \quad O \leq o_{\max}, \end{aligned} \quad (17)$$

where Ω denotes the degree distribution to be optimized, $\tau_{\max}$ is the maximum allowable latency, and $o_{\max}$ is the maximum overhead constraint.



Fig. 4: Two-phase training pipeline of MCRC at each curriculum stage. Phase 1 (Imitation/Distillation): at the base stage ($k = 16$) the policy is pretrained by minimizing KL divergence against the ISD expert; at each subsequent stage of the cascade the expert is the extended teacher distribution obtained from the converged policy of the previous code length. The pretrained weights then warm-start Phase 2 (Reinforcement Learning), where the policy is further optimized via A2C with reward feedback from the MC simulator.

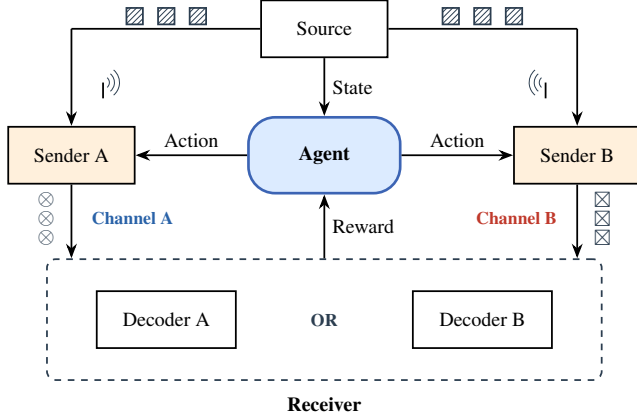


Fig. 5: System model and MCRC training framework. The RL agent observes the system state from the source, selects the transmission action for Sender A and Sender B, and receives reward feedback from the receiver based on decoding performance over independent erasure channels.

B. System Model and MCRC Framework

Consider an MC transmission system designed for latency-sensitive industrial applications, as illustrated in Fig. 5. The system consists of a single transmitter and a receiver, connected via M distinct wireless paths.

In this work, rateless codes are used in the *application layer* to shield the differences in the low-level communication, consistent with prior reliable low-latency communication systems [28], [29], [45]. As illustrated in Fig. 6, the same source block is carried over M parallel links, each with its own transmission medium, rather than being jointly coded at the physical layer. Each link independently performs BP decoding on its received symbols, and the source block is declared recovered once any link succeeds. This application-layer realization requires no cross-path alignment or synchronization, which makes it broadly deployable, and is the setting adopted throughout this work.

The source data is first partitioned into k source symbols, which are encoded into a stream of rateless coded symbols using a degree distribution that is first initialized from the ISD via IL, then cascaded across code lengths via curriculum-style transfer, and further optimized through RL, as illustrated

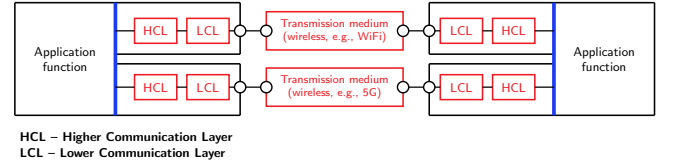


Fig. 6: Rateless codes in the application layer with a multi-connectivity structure [28], [29], [45].

in Fig. 4. Each path uses the shared degree distribution Ω to independently sample and generate its own stream of coded symbols; that is, for each transmission opportunity, path j independently draws a degree $d^{(j)} \sim \Omega$ and generates a distinct coded symbol $x^{(j)}$. This independent sampling ensures that different paths deliver distinct encoded content to the receiver, maximizing coding diversity under the shared degree distribution. Each path independently delivers its encoded packets to the receiver.

We model the M paths as independent binary erasure channels (BECs) with heterogeneous channel quality [35]. Let ε_j denote the packet erasure probability for path j , where $j = 1, \dots, M$. At the receiver, each path maintains its own decoding buffer and performs BP decoding independently. Let $\mathcal{R}_j$ denote the set of successfully received symbols from path j . Each path independently attempts to recover the k source symbols from its own received symbol set.

C. MCRC Algorithm Design

The RL agent, following the deep RL paradigm [31], [46], is formulated as a single-step policy optimization. The state is the source block length:

$$s = k, \quad (18)$$

since the degree distribution must be fixed prior to transmission and only the block length is known at the encoder.

Given $s = k$, the policy network directly outputs a complete degree distribution:

$$\Omega = \pi(k; \theta_\pi) = [\Omega(1), \Omega(2), \dots, \Omega(D_{\max})], \quad (19)$$

where $\sum_{d=1}^{D_{\max}} \Omega(d) = 1$. The degree d_i for each coded symbol is then sampled independently from Ω at transmission time.

There is no per-symbol action selection; the only decision is the choice of Ω conditioned on k .

A sparse reward signal is adopted: the agent receives +1 only when decoding succeeds jointly within the latency constraint *and* within the overhead budget, and -1 otherwise:

$$R = \begin{cases} +1, & \text{if } \beta = 1, \tau < \tau_{\max}, \text{ and } O \leq o_{\max}, \\ -1, & \text{otherwise,} \end{cases} \quad (20)$$

where β denotes the recovery indicator ($\beta = 1$ means all k source symbols are successfully recovered), τ is the end-to-end decoding latency, and $O = (i - k)/k$ is the transmission overhead at the moment of decoding with i coded symbols transmitted. The joint condition enforces that the agent must find a degree distribution that simultaneously satisfies the latency bound $\tau_{\max}$ and the overhead budget $o_{\max}$; violating either constraint yields a negative reward. The recovery indicator β is obtained from the decoding outcome reported by the receiver. As a rateless code, the receiver keeps accumulating coded symbols and attempting BP decoding, returning an ACK once all k source symbols are recovered and treating a timeout as a NACK; during training, the same indicator is provided by the MC simulator.

Decoding is considered successful if any path accumulates sufficient encoded symbols to fully recover the k source symbols within both the latency constraint $\tau_{\max}$ and the overhead constraint $o_{\max}$. This architecture improves reliability through path diversity, since multiple paths independently attempt decoding, increasing the overall probability of successful recovery within the latency and overhead bounds. The complete training procedure is summarized in Algorithm 1.

D. Cross-Path Credit Assignment Analysis

A central difficulty of RL in the MC setting is credit assignment: the reward (20) is a single scalar attached to the *joint* outcome of M parallel decoders, and it is not obvious how this joint feedback should be apportioned to the encoding decisions carried by individual paths. The following proposition shows that, owing to the factorized sampling structure of MCRC, the policy gradient of the joint return decomposes *exactly* into per-path trajectory terms.

Proposition 2 (Per-trajectory decomposition of the joint return): Consider one episode in which path j independently draws degrees $\{d_i^{(j)}\}_i$ i.i.d. from $\Omega(\cdot; \theta_\pi) = \pi(k; \theta_\pi)$, and let R denote the episodic return. Then

$$\nabla_{\theta_\pi} \mathbb{E}[R] = \mathbb{E} \left[(R - V(s; \theta_v)) \sum_{j=1}^M g_j \right], \quad (21)$$

$$g_j = \sum_i \nabla_{\theta_\pi} \log \Omega(d_i^{(j)}; \theta_\pi), \quad (22)$$

where g_j depends only on the degree samples of the single trajectory transmitted on path j . Moreover, under redundant-mode independence,

$$\nabla_{\theta_\pi} \log P_{\text{fail}}^{\text{MP}} = \sum_{j=1}^M \nabla_{\theta_\pi} \log P_{\text{fail}}^{(j)}. \quad (23)$$

Proof: Since the degree samples are drawn independently across paths and symbols, the likelihood of the episode factorizes as $P_{\theta_\pi}(\{d_i^{(j)}\}) = \prod_{j=1}^M \prod_i \Omega(d_i^{(j)}; \theta_\pi)$. Applying the score-function identity $\nabla_{\theta_\pi} \mathbb{E}[R] = \mathbb{E}[R \nabla_{\theta_\pi} \log P_{\theta_\pi}]$, in which any action-independent baseline $V(s; \theta_v)$ leaves the expectation unbiased, and noting that $\nabla_{\theta_\pi} \log P_{\theta_\pi} = \sum_j g_j$, yields (21)–(22). Equation (23) follows by taking the logarithm of (8) and differentiating term by term. ■

Proposition 2 carries two design implications. First, (23) shows that in the log domain the joint objective is *additive* over paths, so each path's marginal influence on the joint failure probability is well defined and separable. Second, (21) shows that the shared-advantage estimator automatically implements this separation: each path contributes exactly its own trajectory score g_j , weighted by the common advantage $R - V(s)$. Cross-path credit assignment is therefore resolved analytically by the factorized likelihood, and no heuristic per-path reward shaping or outcome attribution is required—this justifies the simple sparse joint reward in (20) and the factorized log-likelihood objective used in the training procedure below.

E. Knowledge-Guided Training: Expert Imitation and Cascaded Curriculum Transfer

The training of MCRC follows a knowledge-guided curriculum. Instead of training a policy from random initialization at every code length, MCRC is trained along a doubling code-length ladder $\mathcal{K} = \{k_0, k_1, \dots, k_S\}$ with $k_{m+1} = 2k_m$ (in our experiments, $16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$). At the base stage, the ISD serves as the expert and initializes the policy through IL, as illustrated in Fig. 4; at every subsequent stage, the converged policy of the previous (smaller) code length serves as the *teacher*: it is analytically extended to the new code length by a degree-distribution extension operator, distilled into the student policy, and then fine-tuned by RL in the MC environment shown in Fig. 5. Each stage thus starts from a near-optimal, knowledge-consistent initialization rather than from scratch.

1) Stage-Wise RL Fine-Tuning: Within each curriculum stage, the actor-critic training employs the advantage actor-critic (A2C) framework [42], [43], and entropy regularization with a temperature decay mechanism is applied to encourage exploration [47]. Consistent with Proposition 2, the policy loss aggregates the per-path trajectory scores of (22):

$$\mathcal{L}_\pi(\theta_\pi) = \sum_{j=1}^M \sum_i \log \Omega(d_i^{(j)}; \theta_\pi), \quad (24)$$

$$\nabla_{\theta_\pi} J = \nabla_{\theta_\pi} \mathcal{L}_\pi \cdot A(s) + \lambda_H \nabla_{\theta_\pi} H(\Omega(\cdot; \theta_\pi)), \quad (25)$$

where $\{d_i^{(j)}\}$ are the degree samples drawn from Ω on each path j during the episode, $A(s) = R - V(s; \theta_v)$ is the advantage function computed using the critic's value estimate, and λ_H is the entropy regularization coefficient. By Proposition 2, this factorized objective is not an approximation but the exact score of the joint episode likelihood, so each path receives precisely its own credit under the shared advantage. The critic

Algorithm 1: MCRC Training with Expert Imitation and Cascaded Curriculum Transfer

```

1 // Actor parameters:  $\theta_\pi$ , Critic parameters:  $\theta_v$ 
Input: Code-length ladder  $\mathcal{K} = \{k_0, \dots, k_S\}$  with  $k_{m+1} = 2k_m$ 
2 for  $m = 0$  to  $S$  do
3   if  $m = 0$  then
4     Expert  $\Omega_{\text{exp}} \leftarrow \rho_{k_0}$  (ISD); // expert imitation
5   else
6     Teacher  $\Omega_{k_{m-1}} \leftarrow \pi(k_{m-1}, \theta_\pi)$ ;
7     Expert  $\Omega_{\text{exp}} \leftarrow \tilde{\Omega}_{k_m}$  via Eqs. (29)–(31); // cascaded transfer
8   end
9   Distill: update  $\theta_\pi$  by minimizing  $D_{\text{KL}}(\Omega_{\text{exp}} \| \pi(\cdot | k_m; \theta_\pi))$ ;
10  State  $s \leftarrow k_m$ ;
11  for  $1 \leq \text{episode} \leq T_{\text{max}}^{(m)}$  do
12    Reset gradients  $d\theta_\pi \leftarrow 0$ ,  $d\theta_v \leftarrow 0$ ;
13     $n_{\text{dec}}, i \leftarrow 0$ ;
14    Compute degree distribution  $\Omega \leftarrow \pi(s, \theta_\pi)$ ;
15    Initialize per-path buffers  $\{\mathcal{B}_j\}_{j=1}^M$ ;
16    while  $\tau < \tau_{\text{max}}$  and  $i < i_{\text{max}}$  do
17      for  $j = 1$  to  $M$  do
18        Sample degree  $d_i^{(j)} \sim \Omega$ ;
19        Generate coded symbol  $x_i^{(j)}$  of degree  $d_i^{(j)}$ ;
20        Transmit  $x_i^{(j)}$  over BEC channel  $j$ ;
21        Receive  $y_i^{(j)}$  from channel  $j$ ;
22      end
23      Store  $y_i^{(j)}$  into respective buffer  $\mathcal{B}_j$ ;
24      Perform independent BP decoding on each path;
25      if new source symbol is decoded then
26         $n_{\text{dec}} \leftarrow n_{\text{dec}} + 1$ ;
27      end
28      if  $\beta = 1$  then
29        break;
30      end
31       $i \leftarrow i + 1$ ;
32    end
33    Generate reward  $R$  according to Eq. (20);
34    Entropy regularization:
35     $d\theta_\pi \leftarrow d\theta_\pi + \lambda_H \nabla_{\theta_\pi} H(\pi(s, \theta_\pi))$ ;
36    Policy gradient (per-path scores, Proposition 2):
37     $\mathcal{L}_\pi \leftarrow \sum_{j=1}^M \sum_i \log \Omega(d_i^{(j)}; \theta_\pi)$ ;
38     $d\theta_\pi \leftarrow d\theta_\pi + \nabla_{\theta_\pi} \mathcal{L}_\pi \cdot (R - V(s, \theta_v))$ ;
39    Value gradient:
40     $d\theta_v \leftarrow d\theta_v + \frac{\partial(R - V(s, \theta_v))^2}{\partial \theta_v}$ ;
41    Update  $\theta_\pi$  using  $d\theta_\pi$  and update  $\theta_v$  using  $d\theta_v$ ;
42  end
43 end
44 Joint fine-tuning: repeat the RL loop with  $k$  sampled from the full
   operating range;
Output: Trained policy  $\pi(\cdot, \theta_\pi)$ 

```

parameters θ_v are updated by minimizing the squared temporal difference error:

$$\mathcal{L}_v(\theta_v) = (R - V(s; \theta_v))^2. \quad (26)$$

2) *Expert Imitation at the Base Scale:* At the base stage ($k_0 = 16$), the policy network is pretrained via IL using the ISD as the expert distribution, enabling it to start from a principled degree distribution and accelerate convergence [48]. The IL loss minimizes the Kullback-Leibler (KL) divergence between the policy output and the expert degree distribution:

$$\mathcal{L}_{\text{IL}}(\theta_\pi) = D_{\text{KL}}(\Omega_{\text{ISD}} \| \pi(\cdot | s; \theta_\pi)), \quad (27)$$

$$D_{\text{KL}}(\Omega_{\text{ISD}} \| \pi) = \sum_{d=1}^{D_{\text{max}}} \Omega_{\text{ISD}}(d) \log \frac{\Omega_{\text{ISD}}(d)}{\pi(d | s; \theta_\pi)}, \quad (28)$$

where Ω_{ISD} is the ISD degree distribution used as the expert reference.

3) *Cascaded Transfer Across Code Lengths:* The key ingredient of the cascade is the extension operator that maps a converged degree distribution Ω_{k_m} at code length k_m to an initialization $\tilde{\Omega}_{k_{m+1}}$ at code length $k_{m+1} = 2k_m$. A naive choice is frequency replication—assigning the mass $\Omega_{k_m}(d)$ to degrees $2d - 1$ and $2d$ with halved weight—but this linear dilation doubles every characteristic degree; in particular, it displaces the degree-1 mass that bootstraps BP decoding and violates the soliton scaling laws $\rho(1) = 1/k$ and $\bar{d} = O(\ln k)$. Instead, MCRC transports only the *learned deviation* from domain knowledge, letting the analytic soliton family carry all code-length-dependent structure.

Specifically, the teacher distribution is first decomposed into the ISD carrier and a multiplicative correction:

$$c_m(d) = \ln \frac{\Omega_{k_m}(d) + \delta}{\rho_{k_m}(d)}, \quad d = 1, \dots, k_m, \quad (29)$$

where ρ_k denotes the ISD at code length k in (1) and δ is a small numerical floor. The correction is then re-indexed by the scale-free log-degree coordinate

$$u_m(d) = \frac{\ln d}{\ln k_m} \in [0, 1], \quad (30)$$

which aligns degrees across scales by $d = k^u$, so that structurally corresponding degrees (e.g., the ripple-insurance spike near $d \approx k/S$ in RSD-like shapes) occupy the same coordinate at both code lengths. A monotonicity-preserving piecewise-cubic Hermite interpolant $\hat{c}_m(\cdot)$ is fitted through the points $\{(u_m(d), c_m(d))\}_{d=1}^{k_m}$, and the extended distribution is obtained by re-modulating the ISD carrier at the new code length:

$$\tilde{\Omega}_{k_{m+1}}(d) = \frac{1}{Z_{m+1}} \rho_{k_{m+1}}(d) \exp\left(\hat{c}_m\left(\frac{\ln d}{\ln k_{m+1}}\right)\right), \quad (31)$$

for $d = 1, \dots, k_{m+1}$, where Z_{m+1} is the normalization constant.

This soliton-referenced construction has three desirable properties. (i) *Consistency:* if the teacher equals the ISD ($c_m \equiv 0$), the operator returns exactly the ISD at the new code length—the cascade initialization never degrades below the analytic baseline used for IL. (ii) *Ripple-bootstrap preservation:* the degree-1 mass is governed by the carrier $\rho_k(1) = 1/k$ and thus automatically follows the correct $1/k$ scaling, instead of being displaced to degree 2 as under frequency replication. (iii) *Release-schedule alignment:* because the release probability (3) depends on the degree primarily through its position relative to k , interpolating the correction in the $\ln d / \ln k$ coordinate approximately preserves the expected ripple-addition schedule $\mathbb{E}[A^{(j)}(L)]$ across scales, so the student starts with a ripple profile close to the teacher's.

At each cascade stage, the student network inherits the teacher's weights and is briefly distilled by minimizing

$$\mathcal{L}_{\text{distill}}(\theta_\pi) = D_{\text{KL}}(\tilde{\Omega}_{k_{m+1}} \| \pi(\cdot | k_{m+1}; \theta_\pi)), \quad (32)$$

which takes the same form as the IL loss (27) with the extended teacher distribution in place of the ISD; the IL phase

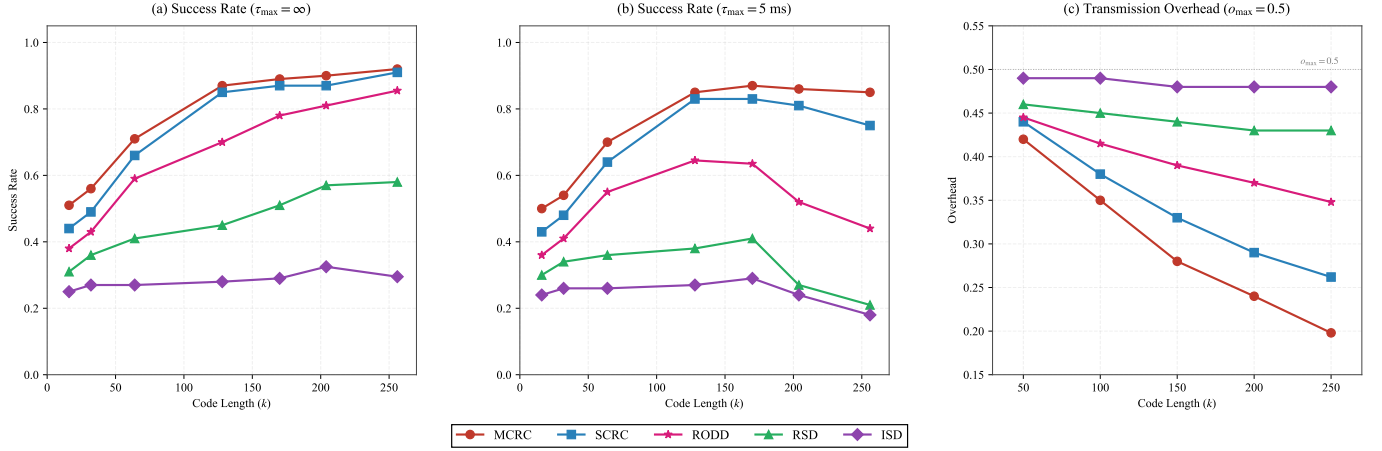


Fig. 7: Comprehensive performance evaluation. (a) Success rate under unbounded latency. (b) Success rate under strict 5 ms latency constraint. (c) Average transmission overhead of successful decodings under $o_{\max} = 0.5$, $\varepsilon_1 = 0.1$, and $\varepsilon_2 = 0.3$. (Legend: MCRC, SCRC [26], RODD [19], RSD [15], ISD [15]).

is thus the $m = 0$ special case of the cascade with the ISD as expert. RL fine-tuning at code length k_{m+1} then refines the distilled policy in the MC environment. After the last ladder stage, a joint fine-tuning phase samples k across the full operating range (16–256) so that the policy also covers intermediate code lengths.

Because every stage starts inside the basin of a near-optimal policy, RL fine-tuning at each scale requires only a fraction of the episodes needed for from-scratch training; aggregated over the ladder, the total training cost is reduced by approximately $3\times$ without loss of final performance, as validated in Section V.

V. SIMULATION RESULTS AND ANALYSIS

A. Decoding Success Rate and Latency Performance Evaluation

We compare the proposed MCRC method against several benchmarks: the classical RSD [15] and ISD [15]; a recent finite-blocklength ripple-oriented degree distribution of rateless codes (RODD) [19]; and a recent single-connectivity rateless code (SCRC) [26]. RODD is a backward, ripple-oriented degree-distribution design tailored to finite block lengths, representing the state of the art among analytically designed distributions. SCRC adopts a recent RL-based method but is confined to a single channel, using the same reward structure as MCRC without multi-path awareness. All methods (RSD, ISD, RODD, and SCRC) are evaluated under the same dual-channel setup ($M = 2$, $\varepsilon_1 = 0.1$, $\varepsilon_2 = 0.3$), ensuring a fair comparison on equal physical resources. The system and training parameters are listed in Table II. Each experiment is repeated 10,000 times, and the results are averaged to ensure statistical reliability.

Regarding the training protocol, the source block length k is the application-facing state variable: the source data is segmented into k symbols according to the service, and k takes values from a discrete, service-determined range (16–256 in our experiments). Training follows the cascaded curriculum

TABLE II: System and Training Hyperparameters

Parameter	Value
<i>System Parameters</i>	
Data packet size	4 KB
Source block size k	16–256
Number of paths M	2
Erasur probabilities ($\varepsilon_1, \varepsilon_2$)	(0.1, 0.3)
Maximum overhead $o_{\max}$	0.5
Latency constraint $\tau_{\max}$	5 ms
<i>Training Hyperparameters</i>	
Hidden layer 1 neurons	128 (ReLU)
Hidden layer 2 neurons	256 (ReLU)
Output activation	Softmax
Optimizer	Adam [49]
Learning rate	1×10^{-5}
Training episodes	20,000–50,000
Code-length ladder $\mathcal{K}$	16 $\rightarrow$ 32 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 256
Distillation steps per stage	1,000
Entropy coefficient λ_H	0.01 $\rightarrow$ 0.001 (decay)

of Section IV-E: the policy is trained along the code-length ladder $\mathcal{K} = \{16, 32, 64, 128, 256\}$, each stage warm-started from the extended teacher of the previous stage, followed by a joint fine-tuning phase in which k is sampled across the full discrete range so that the policy covers all operating code lengths.

As illustrated in Fig. 7, we comprehensively evaluate the models from the dual perspectives of decoding success rate and system latency. Comparing Fig. 7(a) and Fig. 7(b) reveals the profound impact of strict latency boundaries on system performance.

In the unbounded latency scenario ($\tau_{\max} = \infty$), the success rates of all algorithms generally improve as the code length increases, benefiting from the accumulation of redundant information. MCRC consistently achieves the highest success rate, followed by SCRC and RODD, while RSD and ISD exhibit lower performance. Notably, classical degree distribu-

tions (RSD, ISD) show limited success rates even without any latency constraint, particularly at short code lengths ($k \leq 64$), as they are designed under asymptotic assumptions and lack adaptability to finite block lengths and heterogeneous MC conditions.

However, when a strict latency constraint of $\tau_{\max} = 5$ ms is introduced, the performance gap widens further. The success rates of RSD and ISD, as well as RODD, decline significantly when the code length exceeds 128, confirming that degree distributions designed without an explicit latency objective are ill-suited for latency-bounded MC scenarios. In other words, although such finite-blocklength and classical distributions optimize the decoding success rate or coding overhead, they do not explicitly target reliability under a hard latency bound. For example, at a code length of 256, the success rates of RSD and RODD drop sharply. SCRC, benefiting from RL-based optimization, maintains relatively stable performance and demonstrates that learning-based methods can perceive latency constraints during training [31], [46]. However, since SCRC is trained on a single-connectivity environment, its degree distribution is mismatched to the joint channel conditions ($\varepsilon_1, \varepsilon_2$) of the MC setting, limiting its gains compared with MCRC, which is optimized directly under the target MC configuration.

Fig. 7(c) presents the average transmission overhead of successful decodings under $o_{\max} = 0.5$. Since all decodings are capped at $i_{\max} = k(1 + o_{\max})$ symbols, the overhead of each method reflects how efficiently it exploits the symbol budget. The classical RSD and ISD, together with the finite-blocklength RODD, operate relatively close to the overhead cap, indicating that successful decoding often requires nearly all available symbols. In contrast, MCRC achieves the lowest overhead across all code lengths, confirming that multi-path diversity and adaptive degree distribution allow the decoder to succeed well before exhausting the budget. SCRC shows intermediate behavior: its overhead decreases with k but remains higher than MCRC, underscoring the additional efficiency gain from multi-path coordination.

To further investigate the reasons behind the performance variation in the long-code regime, Fig. 8 analyzes the perceived latency distribution across different code lengths.

The results show that RSD and RODD suffer from severe latency accumulation under long code lengths. Specifically, when the code length increases to 256, the latency of RSD rises to over 12 ms and RODD to approximately 8.5 ms, both significantly exceeding the 5 ms threshold, which directly explains their notable decline in decoding success rate under the latency bound. In contrast, SCRC incorporates latency-related rewards during training and can therefore partially control latency growth, resulting in lower latency compared with traditional methods.

Furthermore, the proposed MCRC method leverages the state information from multiple paths to guide the encoding decision process. By dynamically balancing the trade-off between information collection and transmission efficiency, MCRC effectively suppresses latency growth. Even at the maximum code length of 256, MCRC's unconstrained average decoding latency is approximately 6 ms, only slightly

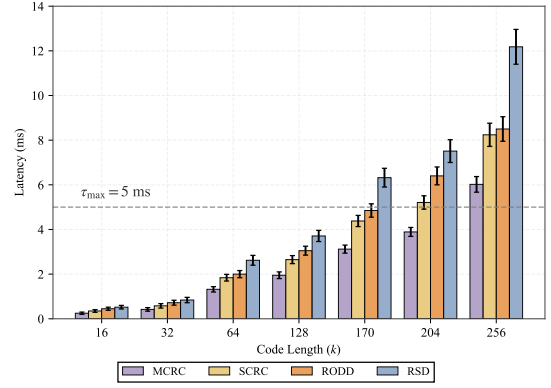


Fig. 8: Average unconstrained decoding latency across different code lengths (latency constraint lifted to expose natural decoding time). The dashed line marks the $\tau_{\max} = 5$ ms bound; methods whose bars exceed this line will exhibit high failure rates in the constrained scenario. ISD is excluded as its decoding latency consistently far exceeds $\tau_{\max}$, making latency statistics uninformative. (Legend: MCRC, SCRC [26], RODD [19], RSD [15]).

above $\tau_{\max}$, indicating that the majority of decoding attempts complete well within the 5 ms window, with a small fraction of failures pulling the unconstrained mean above the bound. This tight latency profile enables MCRC to achieve an 85% decoding success rate at $k = 256$ under $\tau_{\max} = 5$ ms, leading to significant performance gains across all code lengths while maintaining robustness in the long-code regime.

B. Quantitative Decoding and Latency Performance Comparison

We conduct a comparative analysis of four schemes in our experiments: single-path RSD, multi-path RSD, SCRC [26], and the proposed MCRC. The experimental setup is consistent with the parameters described in Table II, with source block size $k = 64$ and 10,000 independent Monte Carlo trials per scheme. Tables III and IV report the aggregate decoding performance and latency statistics across all trials, respectively.

As shown in Table III, under the 5 ms latency constraint, the single-path RSD successfully decodes only 2,831 out of 10,000 source blocks, achieving a success rate of merely 28.3%. Introducing multi-path RSD improves the success rate to 36.2%, demonstrating the benefit of path diversity even with a static degree distribution. SCRC further improves the success rate to 64.1% by adapting to channel conditions through RL optimization. MCRC achieves the best performance with a success rate of 69.7%, confirming that joint multi-path awareness and RL-based degree distribution optimization yield the highest decoding reliability.

From Table IV, we observe that the key difference lies in both average and maximum latency. Classical methods and SCRC frequently exhaust the latency budget ($\tau_{\max} = 5$ ms), causing failure trials to push the maximum latency well above 5 ms (6.80 ms and 5.45 ms for multi-path RSD and SCRC, respectively). In contrast, MCRC learns to decode

TABLE III: Block Decoding Performance Comparison ($k = 64$, 10,000 trials)

Metric	Single Path	Multi-Path	SCRC	MCRC
Total Blocks	10,000	10,000	10,000	10,000
Decoded (∞)	3,247	4,138	6,583	7,094
Decoded (5 ms)	2,831	3,619	6,412	6,973
Success Rate (5 ms)	28.3%	36.2%	64.1%	69.7%

TABLE IV: Latency Distribution Statistics

Statistic (ms)	Single Path	Multi-Path	SCRC	MCRC
Maximum Latency	8.52	6.80	5.45	4.92
Minimum Latency	0.881	0.865	0.852	0.850
Average Latency	5.520	4.610	2.813	1.947
Latency Std. Dev.	1.680	1.520	1.273	0.518

efficiently within the overhead budget $o_{\max}$, so decoding failures predominantly occur when the coded symbol budget $i_{\max}$ is exhausted, well before the latency deadline. As a result, MCRC achieves a lower average latency of 1.95 ms and a maximum latency of 4.92 ms (strictly below $\tau_{\max}$), with smaller variance (0.52 ms vs. 1.27 ms for SCRC), resulting in more stable and predictable end-to-end performance.

The performance gain of MCRC mainly comes from two aspects:

- **Distribution mismatch under single-connectivity training:** SCRC optimizes the degree distribution for a single erasure probability, so when deployed in an MC setting its learned Ω is mismatched to the joint channel conditions $(\varepsilon_1, \varepsilon_2)$. The optimal degree distribution under MC reception differs from its single-connectivity counterpart, since the overall decoding success probability depends on both paths simultaneously.
- **Joint channel-aware optimization:** MCRC trains the RL agent directly within the MC environment, where the reward reflects the joint decoding outcome across all M paths. This allows the agent to implicitly capture the combined channel characteristics and learn a degree distribution Ω^* specifically optimized for the target joint channel conditions $(\varepsilon_1, \varepsilon_2)$, rather than transferring a single-connectivity policy to a mismatched setting.

Overall, MCRC achieves superior reliability and latency by optimizing the degree distribution directly under the joint MC channel conditions, avoiding the performance degradation caused by the single-to-MC distribution mismatch inherent in SCRC.

C. Effect of Knowledge-Guided Training: Imitation and Cascaded Transfer

To validate the role of the knowledge-guided training strategy, we compare the training convergence of MCRC with and without knowledge-guided initialization, where the latter trains the policy at each code length from random initialization.

As shown in Fig. 9, with knowledge-guided initialization the policy at each stage starts from a near-optimal degree distribution—the ISD expert at the base scale ($k = 16$) and the extended teacher distribution (31) at subsequent scales—enabling

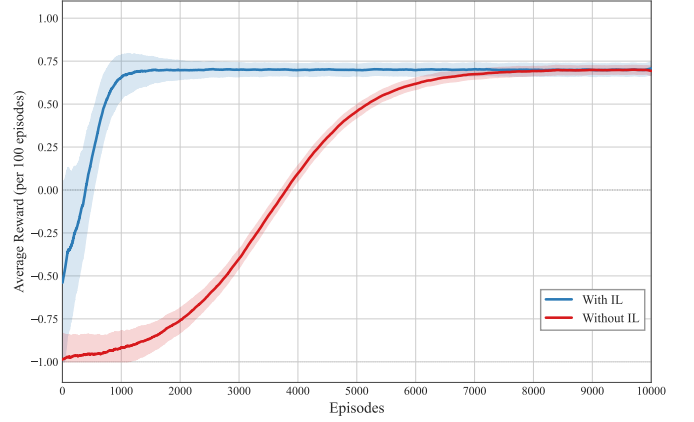


Fig. 9: Training convergence comparison: reward (100-episode moving average) with knowledge-guided initialization (expert imitation plus cascaded transfer) and with random initialization.

the MCRC agent to reach competitive performance within approximately 5,000 episodes. Without knowledge guidance, the agent requires significantly more episodes (approximately $3\times$ longer) to achieve comparable success rates, as it must explore the degree distribution space from a random initialization at every code length. The reward curves in Fig. 9 clearly illustrate the convergence acceleration: the knowledge-initialized agent achieves a stable positive reward within the first 2,000 episodes, while the randomly initialized agent remains in the negative reward region for over 5,000 episodes.

Importantly, the final converged performance of both variants is statistically equivalent, confirming that expert imitation and cascaded transfer serve purely as convergence accelerators without biasing the optimized solution. This is consistent with the consistency property of the extension operator in Section IV-E3: the cascade initialization never falls below the analytic ISD baseline, and RL fine-tuning retains full freedom to depart from the teacher. Aggregated over the code-length ladder, the strategy reduces the offline training cost by approximately $3\times$ while preserving the optimality of the learned distribution, which is desirable for practical deployment.

VI. CONCLUSION

In this paper, we proposed MCRC, a knowledge-guided RL-based degree distribution optimization framework and algorithm design for rateless codes in MC scenarios toward URLLC-oriented IIoT applications [9], [10]. We established, to the best of our knowledge, the first latency-aware MDP formulation for multi-path parallel rateless transmission, derived an upper bound on the multi-path joint decoding failure probability that couples the latency budget, per-path erasure probabilities, and degree distribution, and analytically decomposed the joint return into per-path trajectory terms, providing a theoretical basis for cross-path credit assignment and the sparse joint reward design. Building on this foundation, an MCRC agent was trained with an expert-imitation plus cascaded curriculum transfer strategy, in which policies trained

at small code lengths are analytically extended by a soliton-referenced log-degree interpolation operator and distilled into larger-scale students along a doubling code-length ladder, reducing training cost by approximately $3\times$ without compromising final performance. Simulation results demonstrated that MCRC significantly outperforms classical degree distributions and SCRC approaches [26] in terms of decoding success rate, end-to-end latency, and transmission overhead. Specifically, MCRC achieved a success rate of 69.7% under a strict 5 ms latency constraint at $k = 64$, representing a 41-percentage-point gain over the single-path baseline, and maintained a maximum latency of 4.92 ms with a standard deviation of 0.52 ms, providing effective support for latency-bounded reliability in communications. Future work will explore several promising directions. First, extending the framework to scenarios with more than two paths will further validate the scalability of the approach. Second, we plan to investigate adaptive decoding strategies that dynamically combine received symbols across paths during the BP process, rather than maintaining independent per-path decoders. Such cross-path decoding could exploit partial decoding progress from multiple paths jointly, potentially achieving higher reliability under the same latency budget. Third, while the present solution is obtained by a learning-based method that finds a well-performing policy, an analytical characterization under the joint latency-and-reliability constraints is still lacking; developing such a closed-form analysis is a promising direction for future work.

REFERENCES

- [1] J. Park, S. Samarakoon, H. Shiri, M. K. Abdel-Aziz, T. Nishio, A. Elgabli, and M. Bennis, "Extreme ultra-reliable and low-latency communication," *Nature Electronics*, vol. 5, no. 3, pp. 133–141, 2022.
- [2] C. She, C. Yang, and T. Q. S. Quek, "Radio resource management for ultra-reliable and low-latency communications," *IEEE Communications Magazine*, vol. 55, no. 6, pp. 72–78, 2017.
- [3] A. Nasrallah, A. S. Thyagaturu, Z. Alharbi, C. Wang, X. Shao, M. Reisslein, and H. ElBakoury, "Ultra-low latency (ULL) networks: The IEEE TSN and IETF DetNet standards and related 5G ULL research," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 1, pp. 88–145, 2019.
- [4] Y. Polyanskiy, H. V. Poor, and S. Verdú, "Channel coding rate in the finite blocklength regime," *IEEE Transactions on Information Theory*, vol. 56, no. 5, pp. 2307–2359, 2010.
- [5] G. Durisi, T. Koch, and P. Popovski, "Toward massive, ultra-reliable, and low-latency wireless communication with short packets," *Proceedings of the IEEE*, vol. 104, no. 9, pp. 1711–1726, 2016.
- [6] M.-T. Suer, C. Thein, H. Tchouankem, and L. Wolf, "Multi-connectivity as an enabler for reliable low latency communications—an overview," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 1, pp. 156–169, 2020.
- [7] J. Sachs, G. Wikstrom, T. Dudda, R. Baldemair, and K. Kittichokechai, "5g radio network design for ultra-reliable low-latency communication," *IEEE Network*, vol. 32, no. 2, pp. 24–31, 2018.
- [8] J. Rao and S. Vrzic, "Packet duplication for URLLC in 5G: Architectural enhancements and performance analysis," *IEEE Network*, vol. 32, no. 2, pp. 32–40, 2018.
- [9] E. Sisinni, A. Saifullah, S. Han, U. Jennehag, and M. Gidlund, "Industrial internet of things: Challenges, opportunities, and directions," *IEEE Transactions on Industrial Informatics*, vol. 14, no. 11, pp. 4724–4734, 2018.
- [10] P. Popovski, Č. Stefanović, J. J. Nielsen, E. de Carvalho, M. Angelichinoski, K. F. Trillingsgaard, and A.-S. Bana, "Wireless access in ultra-reliable low-latency communication (URLLC)," *IEEE Transactions on Communications*, vol. 67, no. 8, pp. 5783–5801, 2019.
- [11] D. J. C. MacKay, "Fountain codes," *IEE Proceedings - Communications*, vol. 152, no. 6, pp. 1062–1068, 2005.
- [12] A. Shokrollahi, "Raptor codes," *IEEE Transactions on Information Theory*, vol. 52, no. 6, pp. 2551–2567, 2006.
- [13] J. Castura and Y. Mao, "Rateless coding over fading channels," *IEEE Communications Letters*, vol. 10, no. 1, pp. 46–48, 2006.
- [14] P. Maymounkov and D. Mazières, "Rateless codes and big downloads," in *Peer-to-Peer Systems II (IPTPS 2003)*, ser. Lecture Notes in Computer Science, vol. 2735. Springer, 2003, pp. 247–255.
- [15] M. Luby, "LT codes," in *Proc. 43rd Annu. IEEE Symp. Found. Comput. Sci. (FOCS)*, 2002, pp. 271–280.
- [16] O. Etesami and A. Shokrollahi, "Raptor codes on binary memoryless symmetric channels," *IEEE Transactions on Information Theory*, vol. 52, no. 5, pp. 2033–2051, 2006.
- [17] V. Bioglio, R. Gaeta, and M. Grangetto, "Optimizing the degree distribution of LT codes with an importance sampling approach," *IEEE Communications Letters*, vol. 22, no. 1, pp. 21–24, 2018.
- [18] J. Yin, Y. Fu, J. Tang, and H. Huang, "Optimized degree distribution of rateless codes with lower complexity," *IEEE Communications Letters*, vol. 28, no. 3, pp. 458–462, 2024.
- [19] H. Huang, J. Yin, Y. Fu, and Z. Wang, "Ripple-oriented degree distribution of rateless codes: A backward design approach for finite block lengths," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 11, pp. 17 304–17 320, nov 2025.
- [20] W. Yao, B. Yi, T. Huang, and W. Li, "Poisson robust soliton distribution for LT codes," *IEEE Communications Letters*, vol. 20, no. 8, pp. 1499–1502, 2016.
- [21] J. H. Sørensen, P. Popovski, and J. Østergaard, "Design and analysis of LT codes with decreasing ripple size," *IEEE Transactions on Communications*, vol. 60, no. 11, pp. 3191–3197, 2012.
- [22] J. Guo, D. Zhang, I. Lee, Y. Li, and M. Shirvanimoghaddam, "Partitioned analog fountain codes for short packet communications," *IEEE Communications Letters*, vol. 28, no. 6, pp. 1248–1252, 2024.
- [23] Y. Savchenko and Y. Liu, "Optimizing degree distributions of LT-based codes with deep reinforcement learning," in *IEEE INFOCOM 2019 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2019, pp. 228–233.
- [24] Z. Qin, Z. Fei, J. Huang, X. Wang, M. Xiao, and J. Yuan, "Optimizing distribution and feedback for short LT codes with reinforcement learning," *IEEE Transactions on Communications*, vol. 73, no. 2, pp. 1169–1185, 2025.
- [25] J. Tang, Y. Fu, Y. Qiao, and J. Yin, "Intelligent degree distribution of rateless codes in industrial scenarios," in *2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, 2024, pp. 1–6.
- [26] Z. Wang, Y. Fu, and H. Huang, "Latency-bounded reliability-oriented degree distribution of rateless codes: A knowledge-assisted reinforcement learning method," *IEEE Communications Letters*, vol. 29, no. 10, pp. 2366–2370, oct 2025.
- [27] S. M. M. Shahabi, Y. Zhang, and M. Shikh-Bahaei, "Reinforcement learning-based fountain codes for latency-sensitive transmission," in *2023 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2023, pp. 272–276.
- [28] H. Huang, Y. Fu, and J. Yin, "Latency-reliability tradeoff for timeliness and accuracy of dynamic systems using rateless codes," *IEEE Internet of Things Journal*, vol. 13, no. 5, pp. 8398–8412, mar 2026.
- [29] M. Yuan, Y. Qiao, Y. Fu, and J. Tang, "Fountain codes for reliable and deterministic packet transmission in industrial cloud control systems," *IEEE Internet of Things Journal*, vol. 10, no. 8, pp. 7114–7125, 2023.
- [30] M. Bennis, M. Debbah, and H. V. Poor, "Ultrareliable and low-latency wireless communication: Tail, risk, and scale," *Proceedings of the IEEE*, vol. 106, no. 10, pp. 1834–1853, 2018.
- [31] N. C. Luong, D. T. Hoang, S. Gong, D. Niyato, P. Wang, Y.-C. Liang, and D. I. Kim, "Applications of deep reinforcement learning in communications and networking: A survey," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 4, pp. 3133–3174, 2019.
- [32] A. Feriani and E. Hossain, "Single and multi-agent deep reinforcement learning for AI-enabled wireless networks: A tutorial," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 1226–1252, 2021.
- [33] Q. Zhao, S. Paris, T. Veijalainen, and S. Ali, "Hierarchical multi-objective deep reinforcement learning for packet duplication in multi-connectivity for URLLC," in *2021 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit)*, 2021, pp. 142–147.
- [34] J. Xue, K. Yu, T. Zhang, H. Zhou, L. Zhao, and X. Shen, "Cooperative deep reinforcement learning enabled power allocation for packet duplication URLLC in multi-connectivity vehicular networks," *IEEE Transactions on Mobile Computing*, vol. 23, no. 8, pp. 8143–8157, 2024.
- [35] T. Richardson and R. Urbanke, *Modern Coding Theory*. Cambridge University Press, 2008.

- [36] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 5, pp. 1183–1210, 2021.
- [37] M. Shirvanimoghaddam, Y. Li, and B. Vucetic, "Raptor codes for M2M streaming applications," *IEEE Communications Letters*, vol. 19, no. 4, pp. 557–560, 2015.
- [38] R. Gallager, "Low-density parity-check codes," *IRE Transactions on Information Theory*, vol. 8, no. 1, pp. 21–28, 1962.
- [39] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.
- [40] R. S. Sutton, D. McAllester, S. Singh, and Y. Mansour, "Policy gradient methods for reinforcement learning with function approximation," in *Advances in Neural Information Processing Systems 12 (NeurIPS)*, 1999, pp. 1057–1063.
- [41] R. J. Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," *Machine Learning*, vol. 8, no. 3–4, pp. 229–256, 1992.
- [42] V. R. Konda and J. N. Tsitsiklis, "On actor-critic algorithms," *SIAM Journal on Control and Optimization*, vol. 42, no. 4, pp. 1143–1166, 2003.
- [43] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, "Asynchronous methods for deep reinforcement learning," in *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, ser. JMLR Workshop and Conference Proceedings, vol. 48, 2016, pp. 1928–1937.
- [44] 3GPP, "Study on access traffic steering, switch and splitting support in the 5G System (5GS) architecture," 3rd Generation Partnership Project, Technical Report TR 23.793, 2018, release 16.
- [45] S. Gangakhedkar, H. Cao, A. R. Ali, K. Ganesan, M. Gharba, and J. Eichinger, "Use cases, requirements and challenges of 5g communication for industrial automation," in *2018 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2018, pp. 1–6.
- [46] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [47] E. Jang, S. Gu, and B. Poole, "Categorical reparameterization with Gumbel-Softmax," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Toulon, France, Apr. 2017.
- [48] A. Hussein, M. M. Gaber, E. Elyan, and C. Jayne, "Imitation learning: A survey of learning methods," *ACM Computing Surveys*, vol. 50, no. 2, pp. 1–35, 04 2017.
- [49] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, May 2015.